Phoenix ABS 40-43 · Launch-Day Checklist

Klima D9-5 · aft deploy on motor ejection · flight computer is a logging altimeter (fires nothing) · LaunchControl Pi for the igniter. Tick each box in order. If something is unclear, stand down.

Date: Site: Range Safety Officer:

LaunchControl operator: Wind (m/s) / direction: Fire-ban status checked at:

READINESS GATE. Fly only with a live igniter if all of these are true. If any is false, run the whole sheet as a **dry rehearsal** with a dummy load and no live igniter.

- [ ] Igniter firing circuit is built, and `FIRE_PIN` / `ARM_SENSE_PIN` / `CONTINUITY_PIN` are confirmed against it (LaunchControl README, "Hardware contract").
- [ ] `ignition.py` arm/fire/cutoff has been bench-tested against a dummy load (LED + resistor), with cutoff timing measured.
- [ ] Wi-Fi name/password changed from the defaults on both the flight computer (`config.h`) and the Pi (`config.py`, `network/*.conf`).
- [ ] Built rocket balanced: CG ≤ **226 MM FROM THE TIP**, measured with a loaded D9, the bay packed and the line installed. Add ballast to the EbayCap if needed (~0.6 mm per g).

0 · Go / No-Go: morning of the launch

- [ ] **Fire ban (eldningsförbud)** checked for the municipality today (kommun website, Länsstyrelsen, or the brandrisk / MSB app). A ban means no launch.
- [ ] Wind forecast **≤ 3–4 m/s**. Leaving the 1 m rod at about 11 m/s is below the 15 m/s target, so fly on calm days only. Drift under the canopy is about 5 m/s.
- [ ] Site confirmed: club launch day (SMRK Högsäter), or written landowner OK. If within 10 km of Landvetter or Säve, the airport has been called and has agreed.

1 · Before leaving for the range

- [ ] LaunchControl fired-latch cleared with `scripts/clear_fired_latch.py`, and no "latch already set" warning in the Pi's boot log. **Re-check after any bench test today**: a dummy-load test sets the latch.
- [ ] C1 is fitted on *this* sled (or the transient was bench-verified). Check it, don't rely on memory.
- [ ] Flight-computer LiPo charged, and Pi power source charged.
- [ ] Flight-computer flash has **≥ 5 MB free** (`LOGS` over USB or `GET /logs`). Old logs pulled and backed up, then deleted. Below 5 MB the self-test fails on purpose.
- [ ] Flight-computer battery **unplugged for transport**. There is no power switch, so plugging in the battery powers the bay.

Pack list

- [ ] Rocket: nose, FwdTube/AftTube with sled, fin can, motor cap
- [ ] Klima **D9-5** motors (D9-3 is too early and D9-7 too late; don't swap)
- [ ] Igniters for the D9
- [ ] Canopy ⌀500, 5 mm retention line, Nomex blanket / wadding
- [ ] Tape (one wrap on each motor casing)
- [ ] Hex keys for M5 nose screws and M3 bay screws, spare screws
- [ ] Scale + balance edge for the CG check, ballast
- [ ] Launch pad: hub, 3 legs + hinge pins, thumbscrew, deflector
- [ ] ⌀4 mm steel rod, 1000 mm
- [ ] Blast plate / board, if the ground is not bare
- [ ] Water + fire extinguisher
- [ ] LaunchControl Pi, firing box, leads, arm switch / key
- [ ] Phone for the webapp; laptop + USB-C cable (serial console / log pull)
- [ ] This checklist, pen, spare batteries

2 · At the range: prep area

Ground station

- [ ] LaunchControl Pi powered, phone joined to its hotspot, webapp loads.
- [ ] AP+STA concurrency (`iw phy0 info`) has been confirmed at least once for this Pi/kernel.

Aft section: recovery + motor

- [ ] **1.** Line, canopy and Nomex all fit in the bay by hand. The line runs through the bulkhead eyelet and the fin-can staple lug (no Kevlar leaders).
- [ ] **2.** Pack **forward to aft**: canopy against the bulkhead, then the line, then the **Nomex/wadding over the fin-can gas port**. Nomex is **mandatory**: the charge blows straight into the bay. A nylon/polyester line needs a Nomex sleeve at the staple end.
- [ ] **3.** Motor: **one wrap of tape** on the D9-5 casing (seals the gap that lets gas leak past to the nozzle), push it in against the thrust ring, then screw on the cap **hand-tight + witness mark**. The cap carries the ejection load.
- [ ] **4.** Fin can: push in, turn until the **tab drops into the notch**, push until **2 clicks**. Pull gently to check it holds.
- [ ] **5.** **About a 5 mm gap at the joint = wrong clocking. Do not fly.** No clicks = do not fly. Never tape over the latch tongues (they vent the bay).

Forward section: avionics

- [ ] **1.** Nose off. Plug in the flight-computer battery. Listen for the **self-test pass tone**. A fail tone means stand down and check over USB with `STATE`.
- [ ] **2.** Nose back on with the 3 × M5 screws. From here on, **handle gently**: a jolt above **3 g held for 100 ms** counts as a launch.
- [ ] **3.** LaunchControl status view shows the flight computer as reachable (`/status`: `logger_ready` true, `log_free_bytes` OK).

3 · Pad setup + igniter

- [ ] **1.** Pad on bare soil, gravel or a blast plate, never dry grass. Water + extinguisher at the pad. Spectators and cars upwind.
- [ ] **2.** Rod in the hub: clamp **snug, not over-tight**, rod square to the hub (1° of tilt is ~17.5 mm of wander at the tip). Deflector fitted.
- [ ] **3.** Carry the rocket to the pad and slide the lug onto the rod. It must slide freely.
- [ ] **4.** Rocket vertical, undisturbed.
- [ ] **5.** Igniter installed per Klima instructions, with the **firing-circuit arm switch OPEN**.
- [ ] **6.** Webapp **continuity reads OK**. If it doesn't, check the igniter connection. **Never bypass the continuity gate.**
- [ ] **7.** Webapp **arm-sense reads OPEN** while the switch is open. A wrong reading means the UI can't be trusted: stand down.
- [ ] **8.** **Do not handle, bump or jar the rocket from here on.**

STAND-DOWN / RESET. If the rocket is bumped hard, comes off the rod, or the result of `/arm` is genuinely unclear: open the arm switch, remove the igniter, **power-cycle the sled** (unplug and re-plug the battery, listen for the pass tone), then restart from §3 step 3.

4 · Wireless arm (re-zero the flight computer)

- [ ] **1.** Rocket has been vertical and undisturbed since §3 step 4. `/arm` takes a new pad zero (baro + accel baseline), so the rocket must be in its final position.
- [ ] **2.** Trigger **/arm** from the LaunchControl webapp.
- [ ] **3.** Expect the connection to **drop about 0.5 s after success**. An "ambiguous" result plus a dropped connection most likely means it **worked**.
- [ ] **4.** **Do not retry /arm.** A successful arm turns the radio off until the next power-cycle. If the outcome is genuinely unclear, use the stand-down box above.
- [ ] **5.** Confirm arm success from the last status the webapp captured. Don't proceed on "probably fine".

5 · Final arm + fire (igniter channel)

- [ ] **1.** All personnel at the range's safe distance. RSO gives the range clear.
- [ ] **2.** Designated operator only closes the **physical arm switch**.
- [ ] **3.** Webapp **arm** succeeds (needs arm-sense closed **and** continuity OK). If it fails, stand down and fix the igniter circuit. Never fire on a failed arm.
- [ ] **4.** **Arm window is 15 s.** If the hold runs longer, **re-arm**. Don't just press fire again.
- [ ] **5.** Countdown per range procedure, then **FIRE**.
- [ ] **6.** No ignition within a few seconds = **misfire**. The fire channel is one-shot. Open the arm switch and follow the range misfire wait before approaching the pad.

6 · Flight, recovery + data

- [ ] **1.** Watch the flight. Note the apogee time and when ejection happens (video if possible).
- [ ] **2.** Recover. Check the motor is cool. Note damage to the fin can, latches, line and canopy.
- [ ] **3.** Open the arm switch on the firing box. The LaunchControl fired-latch now reads "set". **That is expected.** Clear it (`scripts/clear_fired_latch.py`) only before the next flight.
- [ ] **4.** **Pull the log.** Power-cycle the sled (the Wi-Fi turned itself off at launch). **The restart starts a new, empty log, so the flight is usually the <u>second-highest</u> number.**
- [ ] **5.** Wi-Fi: join the flight computer's network, `curl http://192.168.4.1/logs`, then `curl -OJ "http://192.168.4.1/log?n=NN"`. Or USB: `LOGS`, then `DUMP NN` (save what's between BEGIN/END).
- [ ] **6.** Back up the CSV **in two places** (laptop + phone/cloud) and check that it opens. Only then free the space with `DEL NN`.
- [ ] **7.** In the CSV, find `APOGEE_CONFIRMED` / `APOGEE_TIMEOUT` and the ejection accel spike. Expected: ejection **~0.5 s after apogee**. A large gap means the motor delay is wrong for this rocket. Fix it before the next flight.
- [ ] **8.** Unplug the flight-computer battery before transport.

Quick reference

Flight computer	
Address	`http://192.168.4.1`
Wi-Fi	`WIFI_AP_SSID` / `…PASSWORD` in `config.h`; 1 client max
HTTP	`/status` `/config` `/logs` `/log?n=` · POST `/arm`
USB serial	115200 · `STATE` `LOGS` `DUMP n` `DEL n` `FORMAT YES` (wipes all)
Launch detect	3.0 g held 100 ms; turns Wi-Fi off
Log file	`FLIGHTnnn.CSV`, one per power-on
Firmware	`pio run -t upload` keeps logs; `-t erase` wipes them

Expected flight (OpenRocket, D9-5, no wind)	
Liftoff mass	≈ 263 g
Stability	1.92 cal liftoff / 2.14 burnout
Apogee	≈ 172 m at 6.8 s
Max velocity	≈ 50 m/s
Ejection	+0.47 s after apogee, 4.6 m/s
Descent	≈ 5.2 m/s on ø500 canopy
Igniter arm window	15 s

Flight log

#	Time	Motor	Wind m/s	Log file n	Ejection vs apogee	Notes / damage

Compiled from `Electronics/LaunchControl/pad_checklist.md`, `Electronics/Avionics/avionics_build_checklist.md`, `PhoenixABS40-43/Models/AFT_DEPLOY_D9_*.md` (v1–v11), `LAUNCH_SITE.md`, `LaunchPad/Design-Notes-LaunchPad.md`. CO2 / burn-wire steps omitted: Phoenix has no CO2 module (2026-10-19). If this sheet and those documents disagree, the documents win.